Dataset Description

Project Title

StockSense AI: Intelligent Demand Forecasting and Inventory Management System

Dataset Name

Retail Store Inventory Dataset

Dataset Source

Kaggle

Dataset Description

The **Retail Store Inventory Dataset** contains historical data from retail store operations. It includes details about product sales, inventory levels, pricing, demand forecasts, seasonal factors, weather conditions, and promotional activities. This dataset reflects real-world retail scenarios and helps analyze sales trends and inventory performance.

For the **StockSense AI** project, this dataset supports the development of a machine learning model that predicts future product demand. By examining past sales and inventory information, the system assists retailers in maintaining optimal stock levels, minimizing excess inventory, preventing stock shortages, and improving inventory planning.

Dataset Information

Number of Instances: 73,100

Number of Features: 15

Features

- Date
- Store ID
- Product ID
- Category
- Region
- Inventory Level
- Units Sold
- Units Ordered
- Demand Forecast
- Price
- Discount
- Weather Condition
- Holiday/Promotion
- Competitor Pricing
- Seasonality

Input Variables (X)

- Date
- Store ID
- Product ID
- Category
- Region
- Inventory Level
- Units Sold
- Units Ordered
- Price
- Discount
- Weather Condition
- Holiday/Promotion
- Competitor Pricing
- Seasonality

Output Variable (Y)

- Demand Forecast

Purpose of the Dataset

The main purpose of this dataset is to train and evaluate a machine learning model for demand forecasting. The model learns from historical retail data to predict future product demand. This enables businesses to optimize inventory levels, improve stock availability, reduce operational costs, and make informed decisions about inventory management.

Expected Output

The developed system predicts future product demand using historical retail data. These predictions assist with inventory management by helping retailers plan stock levels effectively, minimize overstock and shortages, and make data-driven business choices.